INCH-POUND

MIL-I-27188B (USAF)
30 April 1994
SUPERSEDING
MIL-I-27188A (USAF)
8 NOVEMBER 1967

MILITARY SPECIFICATIONS

INDICATOR, PRESSURE EGU-2/A, OIL, 0-100 PSI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of hermetically sealed, remote indicating, synchro-style, integrally lighted, oil pressure indicator, designated EGU-2/A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

PPP-B-636	Box, Shipping, Fiberboard
PPP-C-795	Cushioning Material Flexible, Cellular, Plastic Film, For Packaging Application

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, Tinker AFB, OK 73145-5990 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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Military

MIL-P-116	Preservation, Method Of
MIL-B-121	Barrier Material, Greaseproofed Waterproofed, Flexible
MIL-C-5541	Chemical Films And Chemical Film Materials For Aluminum And Aluminum Alloys
MIL-T-5796	Transmitter, Pressure, Oil, 0-100 PSI
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-S-20708	Synchros, General Specification For
MIL-T-25624	Transmitter, Pressure, Oil, Engine Mounted 0-100 psi, Type TRU-80/A
MIL-L-27160	Lighting, Instrument, Integral, White General Specification For

STANDARDS

FED-STD-101	Preservation, Packaging And Packing Materials: Test Procedures
FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment and Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristing Measurement Of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-785	Requirements For Reliability Program (For Systems and Equipment)
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MIL-STD-970	Standards & Specifications, Order Of Preference For The Selection Of
MIL-STD-2073/1A	DOD Materiel Procedures For Development & Application Of Packaging Requirements
MS28005	Transmitter, Pressure, Synchro, Aircraft
MS28010	Indicator, Pressure-Synchro, Single, 2-Inch Size
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of

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MS33678 Connector, Receptacle, Electric, Integral Mounting

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave, Building #4, Section D, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3)

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.3 Material

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be sued for all parts of the indicator, except where magnetic materials are essential.

3.3.2 Nonferrous materials. Nonferrous materials shall be used for all parts of the indicator, except where ferrous materials are essential.

3.3.2.1 Recovered materials. Recovered materials are used to the maximum extent possible without jeopardizing the end use of the item.

3.3.3 Metals. Metals shall be corrosion-resident or suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service.

3.3.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.5 Toxic and corrosive fumes. The indicator shall be of such material and construction that fumes, when burned, or gases that combine with the atmosphere to form acid or corrosive alkali will not be liberated.

3.3.6 Protective treatment. When materials are used in the construction of the indicator that are subject to deteriorations when exposed to climatic environmental likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance

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with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction. The indicator shall be designed for use with a TRU-80/A transmitter in accordance with MIL-T-25624 or an MS28005-2 transmitter conforming to MIL-T-5796

3.4.1 Case. The case shall be hermetically scaled and the dimensions shall conform to MS28010.

3.4.1.1 Cap. A separate nonhermetically sealed cap may be used on the back of the case for installing the electrical receptacle. The cap shall be made of nonferrous, low-density metal.

3.4.1.2 Case to cover glass distance. The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed 1/32 inch.

3.4.1.3 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain no more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi. Other concentrations of filling medium shall not be used except with prior approval of the procuring activity.

3.4.2 Lighting. The indicator shall be integrally lighted and shall meet all requirements specified herein. The indicator shall be lighted in accordance with MIL-L-27160.

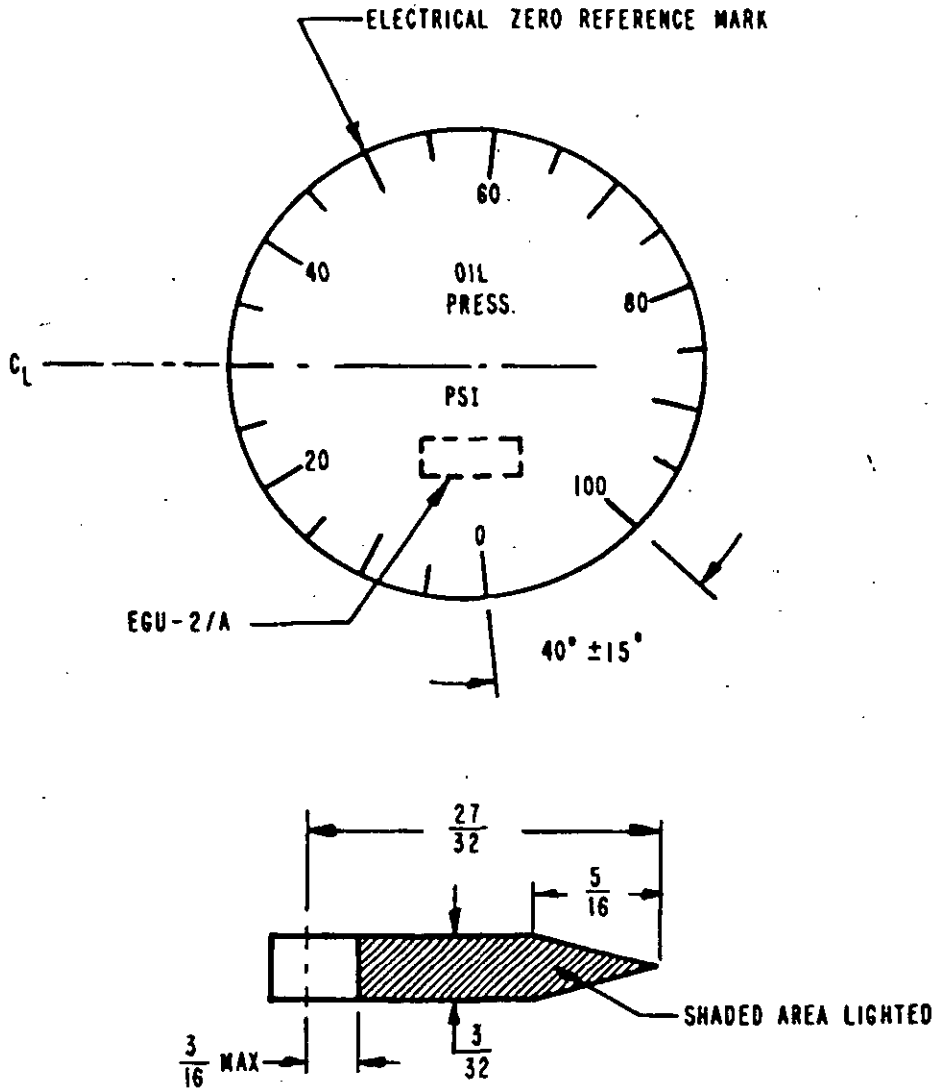
3.4.3 Pointer. The pointer shall conform to figure 1 and shall be suitable for integral lighting. The length shall be such that the tip of the pointer will extend into the scale a distance equal to one- to two-thirds the length of the shortest graduation.

3.4.3.1 Pointer finish. The lighted portion of the pointer shall be finished in accordance with FED-STD-595, color No. 37875.

3.4.4 Dial. The dial shall conform to figure 1. A minimum diameter of 1-3/4 inches shall be maintained across the outside ends of the graduations. The 30-psi graduation shall be located on the horizontal centerline at the left side of the dial.

3.4.4.1 Dial markings. The following markings shall be finished in durable white, color No. 37875 of FED-STD-595. The dimensions of the markings shall be as specified.

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DIMENSIONS IN INCHES

UNLESS OTHERWISE SPECIFIED, TOLERANCES: FRACTIONS ±1/64

FIGURE 1. Dial and Pointer.

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	Height or length Inch ±1/64	Width of line or graduation Inch ± 0.005
Numerals 0, 20, 40, 60, 80, and 100	5/32	0.025
10-pound graduations	3/16	0.031
Graduation at 50 psi	9/64	0.031
Lettering OIL PRESS	1/8	0.020
Lettering PSI	1/8	0.020
5-Pound graduations	3/32	0.020

3.4.4.2 Black markings. The marking EGU-2/A shall be permanently and legibly marked on the dial in letters 1/16 inch high. This marking and the color of all other markings not otherwise specified shall be finished in durable dull black, color No. 37038 of FED-STD-595.

3.4.4.3 Dial visibility. Portions of the minor graduations on the dial shall be visible when viewed from any position within the frustum of a cone the side of which intersects the outer edge of the cover glass and which makes an angle of 30° with the perpendicular through the center of the dial.

3.4.4.4 Dial to cover glass distance. The distance between the indicator dial and the inside of the cover glass shall be the minimum practicable and shall not exceed 0.188 inch.

3.4.5 Filler tube. A tube or other facility for evacuating and filling the case shall be provided at the rear of the case and shall be recessed beneath the level of the back plate surface. If a tube is used, it shall be of such length as to permit three refillings of the case.

3.4.6 Cover glass. The cover glass shall be clear, flat, and free from flaws that would interfere with the normal reading of the instrument. The thickness of the cover glass shall be the minimum practicable and entirely suitable for the purpose.

3.4.7 Fasteners. The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. If screws are used for attaching the dial, they shall be so located that they will not interfere with dial markings or prevent the markings from being located in the proper position.

3.4.8 Electrical zero reference mark. The dial shall be provided with an electrical zero reference mark on the scale portion equidistant in angular degrees between the first and last dial graduations.

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3.4.9 Electrical connector. The indicator shall be provided with an MS33678-14S-5P electrical connector which may be either hermetically or nonhermetically sealed. When the connector is nonhermetically sealed, a hermetically sealed baffle plate shall be used to accomplish sealing of the indicator.

3.4.10 Electrical wiring. The internal and external wiring of the indicator shall be in accordance with figure 2.

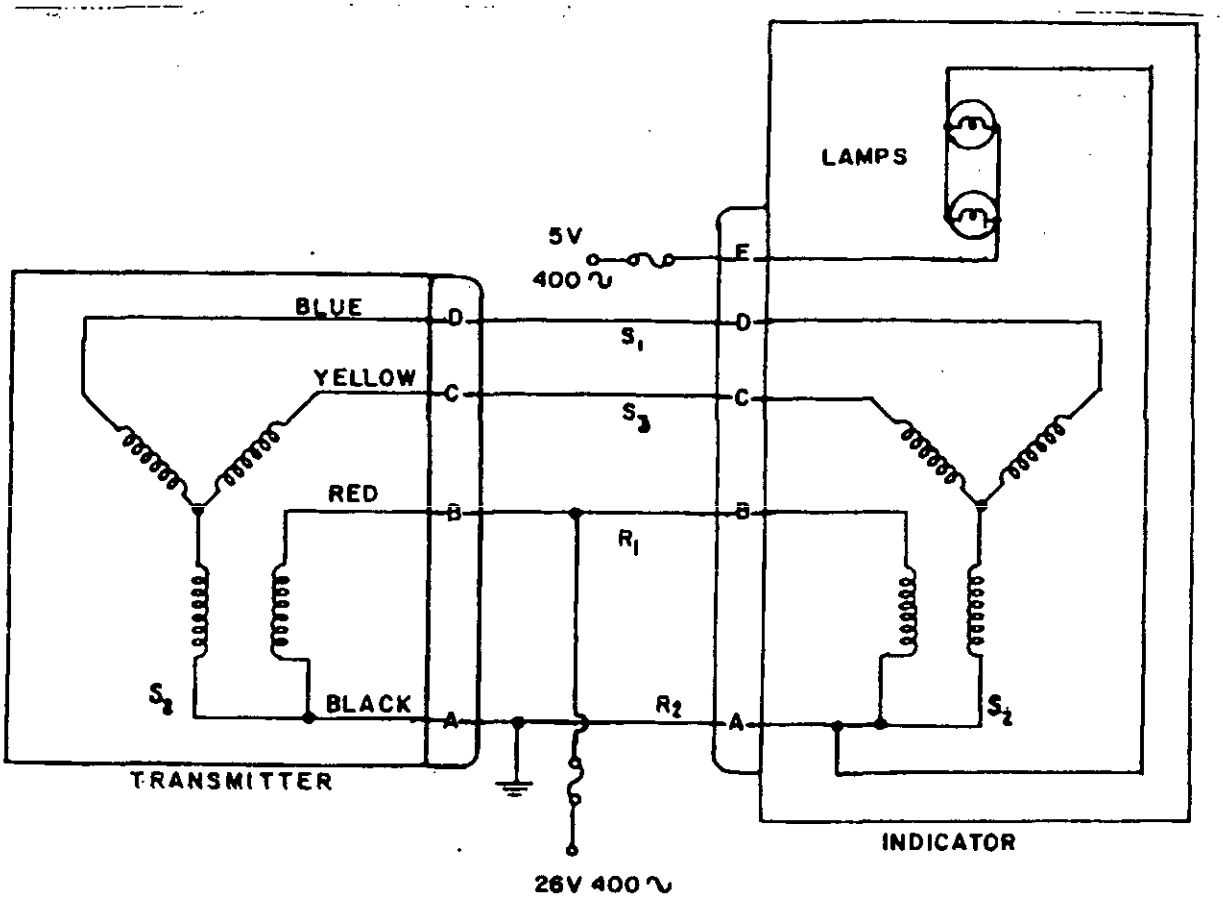


FIGURE 2. Wiring diagram

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3.4.11 Indicator synchro. The indicator synchro shall consist of a salient two-pole wound rotor used as the primary and shall be designed for 26V, 400-cycle, as required, single-phase power, with wye-connected stators as the secondaries. The stators shall be so wound that the amplitudes of the induced voltages measured between any two stator leads will vary approximately sinusoidally with the angular positions of the rotor. Pivots, bearings, and gears shall not bind nor shake and shall be as nearly frictionless as possible.

3.4.11.1 The synchro receiver used shall have electrical characteristics compatible with those defined in MIL-S-20708.

3.4.12 Secondary voltage. The maximum open-circuit voltage measured between any two secondary leads shall be 11.8V ±3 percent.

3.4.13 Radio noise interference. The indicator shall meet the radiated and conducted radio noise interference requirements of MIL-STD-461 and MIL-STD-462.

3.4.14 Reliability program. The contractor shall establish a reliability program in accordance with MIL-STD-785.

3.4.14.1 Reliability. The indicator shall have a minimum mean-time-between-failures (MTBF) of 2,000 hours based upon a discrimination ratio of 1.5 and a confidence factor of 90 percent.

3.4.15 Longevity. The indicator shall have an operating life span of 3,000 hours.

3.5 Performance. The indicator shall operate satisfactorily after subjection to the following conditions:

a. Temperatures -- temperatures ranging from -65° to +71°C

b. Vibration -- vibration of 2g's from 5 to 2000 cps

c. Temperature-altitude -- pressure altitude of 100,000 feet at -55°C

d. Thermal shock -- eight cycles of alternate immersions in tap water at 85° ±2°C

e. Salt fog -- exposure to salt sea atmosphere for 48 hours without evidence of corrosion or deterioration.

3.5.1 Scale error. Scale error shall not exceed the tolerances specified in tables I and II.

3.5.2 Friction error. Friction error shall not exceed the tolerances specified in table II.

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3.5.3. Position error. Position error shall not exceed the tolerances specified in table II.

TABLE I. Scale Error

PSI	Tolerance ±PSI
0	1.0
20	1.0
40	1.0
60	1.0
80	1.0
100	1.0

TABLE II. Test Tolerances

Test	Tolerance ±PSI
Position error at midscale	1.0
Friction	
Room temperature	1.0
High temperature	1.0
Low temperature	1.0
Vibration	
Pointer oscillation	1.0
Pointer variation	1.0
Scale error	
Low temperature	1.0
High temperature	1.0

3.5.4 Dielectric strength. The indicator shall withstand 200V rms at a commercial frequency for a period of 5 seconds.

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3.5.5 Case leakage. Case leakage shall not exceed 0.1 micron cubic foot per hour.

3.5.6 Fogging. There shall be no evidence of moisture or oil fogging of the cover glass when indicator is subjected to a temperature of 85° ±5°C for a minimum period of 2 hours and then the face of the cover glass immediately immersed in an ice bath for 3 minutes.

3.5.7 Effect of lead length. When the lead length between the indicator and transmitter is changed from 5 to 500 feet, the indication shall not change more than 1° of arc. The ground lead shall be kept short.

3.5.8 Magnetic effect. The compass deflection shall not exceed 3° of arc when the indicator is positioned 5-1/2 inches from the compass and the horizontal magnetic field intensity at the indicator is 0.17 to 0.19 oersted.

3.5.9 Unit torque gradient. The unit torque gradient shall be a minimum of 2,000 milligram-millimeters per angular degree of rotation.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 Numerals and letters. Numerals and letters shall conform to MS33558.

3.8 Finishes and protective coatings.

3.8.1 Aluminum alloy parts. Aluminum alloy parts shall be coated with an anodic film conforming to MIL-A-8625 except that small holes, straight threads, case inserts, and parts in hermetically sealed enclosures need not be anodized.

3.8.1.1 Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film conforming to MIL-C-5541.

3.8.1.2 When abraision resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used.

3.9 Weight. The weight of the indicator shall not exceed 0.60 pound.

3.10 Screw threads. Unless otherwise specified, the threads of all machines screws shall conform to MIL-S-7742.

3.11 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

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3.11.1 Nameplate. A nameplate shall be permanently attached to the rear of the case. Where applicable, the information marked thereon shall conform to MIL-STD-130. The nameplate shall not be glued or cemented to the case nor shall metal decals be used.

3.12 Workmanship. The indicator shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, welding, riveting, brazing, plating, painting, machine screw assemblies, and freedom of parts from burrs and sharp edges. Gears shall be of high quality and shall provide a minimum of backlash necessary for smooth operation. Where necessary, antibacklash techniques shall be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted tot he Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of know defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4)

b. Quality conformance inspection (see 4.3)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the inspection conditions specified in paragraph 4.3.1 in the specification.

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4.3.1 Standard atmospheric conditions. Wherever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Attitude. Unless otherwise specified, the indicator shall be tested in the normal operating position with the plane of the dial vertical.

4.3.3 Electrical power. Unless otherwise specified the indicator shall be tested with 26 ±1V, single-phase, 400 ±5-cps a-c power and with 4.5 ±0.5V ac or dc applied to the lighting circuit.

4.3.4. Tapping. Unless otherwise specified, the indicator shall be lightly tapped or vibrated before each reading.

4.3.5 Integral lighting. The integral lighting tests shall be conducted in accordance with MIL-L-27160 in the sequence specified.

4.3.6 Test tolerances. The indicator shall be operated within the tolerances specified in tables I and II.

4.4. Qualification inspection.

4.4.1 Test samples. The test samples shall consist of 10 indicators representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the qualifying activity. Seven of the samples shall be subjected to the reliability test only and the other three shall be subjected to all the tests except reliability.

4.4.2 Qualification conformance inspection. The qualification tests shall consist of all the tests specified under 4.6.

4.4.2.1 Nonstandard parts data approval. When nonstandard electronic parts are used in the construction of the indicator, nonstandard parts data approval shall be a prerequisite to qualification approval.

4.5 Acceptance tests. Acceptance tests shall consists of:

a. Individual tests

b. Sampling plans and tests.

4.5.1 Individual tests. Each indicator shall be subjected tot he following tests as described under 4.6:

a. Examination of product

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b. Individual tests of MIL-L-27160

c. Dielectric strength

d. Electrical zero

e. Scale error and friction at room temperature

f. Position error

g. Sealing

4.5.2 Sampling tests

4.5.2.1 Sampling plan A. One indicator shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests

b. Sampling plan A tests of MIL-L-27160

c. Vibration

d. Low temperature

e. High temperature

f. Fogging

4.5.2.2 Sampling plan B. Unless otherwise specified, three indicators shall be selected at random from the first 15 on the contractor or order and subjected tot he following tests as described under 4.6:

a. Sampling plan A tests

b. Sampling plan B tests of MIL-L-27160

c. Secondary voltage

d. Magnetic effect

e. Effect of lead length

f. Temperature-altitude

g. Low temperature exposure

h. High temperature exposure

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i. Thermal shock

j. Salt fog

k. Radio noise interference

l. Unit torque gradient.

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of the failure have been determined and appropriately corrected.

4.5.2.4 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.6 Test methods

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and markings. Prior to assembly, dials, pointers, and synchros shall be inspected as to markings, dimensions and finishes.

4.6.2. Dielectric strength. A potential of 200V at a commercial frequency shall be applied between any pin of the electrical receptacle and the case of the indicator for a minimum period of 5 seconds. There shall be no breakdown of the insulation.

4.6.3 Electrical zero. The electrical zero test and the verification of leads shall be accomplished on the completed indicator with the case removed. A voltage of 26 ±1V single phase shall be applied to the rotor of the synchro with leads designated R_1 and R_2. R_2 shall be connected to the ground or low potential side of the power input. With a lead designated S_2 connected to R_2, the following conditions shall exist at electrical zero:

a. The voltage between the remaining stator terminals designated S_1 and S_3 shall be a minimum

b. The voltage between R_1 and S_3 shall be near the low null and shall reach the low null when the rotor is rotated clockwise approximately 30° from electrical zero.

4.6.4 Scale error and friction. The indicator shall be tested for scale error at ambient temperature at the test points specified in table I. The indicator shall be lightly tapped or vibrated before each reading to

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remove normal instrument friction. The scale errors shall not exceed the tolerances specified in table I. The friction error test may be conducted in conjunction with the scale error tests. The difference between the readings before and after the indicator is tapped or vibrated shall constitute the friction error. The friction error shall not exceed the tolerances specified in table II. This procedure shall be accomplished both up-scale and down-scale at the test points specified.

4.6.5 Position error. The indicator pointer shall be positioned at the electrical zero reference mark with power applied. The change pointer position when the indicator is held in any position shall not exceed the value specified in table II.

4.6.6 Sealing. The indicator case shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere.

4.6.7 Vibration. The indicator shall be subjected to a vibration test in accordance with MIL-STD-810, method 514, equipment category 6.1, procedure I, curve B. A synchro signal shall be maintained on the stators of the indicator synchro to provide an indication of 45 psi. Pointer oscillation shall not exceed 5 psi double amplitude. Upon completion of this test, the indicator shall be subjected to the scale error test at room temperature. The scale errors shall not exceed the tolerances specified in table I.

4.6.8 Low temperature. The indicator shall be subjected to a temperature of -55° ±2°C for a period of 4 hours. While at this temperature, it shall be checked for friction and scale errors and shall not exceed those specified in table II. In every case, the indicator shall not require more than 3 seconds to come within the tolerance specified.

4.6.9 High temperature. The indicator shall be subjected to a temperature of 71°C for a period of 4 hours. While at this temperature, it shall be checked for scale errors. The scale error and friction tolerances shall not exceed the values specified in table II. The response time shall not exceed 3 seconds.

4.6.10 Fogging. The indicator shall be placed in a controlled ambient temperature of 85° ±5°C for a minimum period of 2 hours. Within 10 seconds after removal from the oven, the indicator shall be immersed to a depth of 1/4 inch in an ice water bath maintained at 0° to 7°C. After 3 minutes, the indicator shall be removed from the fluid bath and quickly wiped dry (compressed air shall not be used). The cover glass and dial shall immediately be inspected and any evidence of fogging or damage shall be cause for rejection.

4.6.11 Secondary voltage. The maximum open-circuit voltage shall be measured between any two secondary leads and shall be 11.8V ±3

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percent. The variation between the three maximum secondary voltages on the indicator synchro as measured between any two stator leads shall not exceed 0.2V.

4.6.12 Magnetic effect. The indicator, first, not operating and second, operating on rated voltage, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. Deflection of the compass at any of the specified positions shall not exceed 3°.

4.6.13 Effect of lead length. The indicator shall not change more than 1° of arc when the lead length between the indicator and transmitter is changed from 5 to 500 feet of the No. 20 AWG gage wire. Equivalent resistances may be substituted for the added wire length. The ground lead shall be kept short.

4.6.14 Temperature-altitude. The temperature-altitude test shall be conducted in accordance with MIL-STD-810, method 504, procedure I (except the temperature shall be -55°C) with a chamber pressure equivalent to an altitude of 100,000 feet. The rate of pressure change shall not exceed an average of 3 psi per second while under these conditions and at the end of a 1-hour period, a scale and friction error test shall be conducted. The errors shall not exceed the tolerances specified in table II. This test may be conducted concurrently with the low temperature exposure test.

4.6.15 Low temperature exposure. The indicator shall be subjected to a temperature of -65° ±2°C for a period of 48 hours. While at this temperature, the indicator shall be operated to assure that the pointer moves over the entire scale range without exhibiting sticky or erratic characteristics. The temperature shall then be raised to -55° ±2°C and held for a period of 24 hours. While at this temperature, the indicator shall be subjected to the scale error and friction test, except that in every case, the indicator shall not require more than 4 seconds to come within the tolerances of table II. The scale error and friction shall not exceed to tolerances specified in table II. All parts of the indicator shall then be maintained at room temperature for a period of not less than 4 hours and the indicator again subjected to the scale error and friction test. The scale errors shall not exceed the tolerances specified in table II. There shall be no damage nor deterioration that will affect subsequent operation.

4.6.16 High temperature exposure. The indicator shall be subjected to a temperature of 71°C for a period of 24 hours and then tested at this

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temperature of scale errors as specified in 4.6.4. The scale error tolerances and friction shall not exceed the tolerances specified in table II. All parts of the indicator shall then be maintained at room temperature for a period of not less than 4 hours and the indicator subjected to the scale error and friction test. The scale errors observed shall not exceed the tolerances specified in table II. There shall be no damage or deterioration that might affect subsequent operation.

4.6.17 Thermal shock. The indicator shall be subjected to eight cycles of immersion in water at a temperature of 85° ±2°C and then in water at a temperature of 5° ±2°C without evidence of moisture penetration or damage to the coating or the case. The length of time for each immersion shall be 30 minutes , with not more than 5 seconds between immersions. After eight continuous cycles have been completed, the indicator shall be subjected to the test specified in 4.6.6.

4.6.18 Salt fog. The indicator shall be subjected to a salt fog test for a period of 48 hours in accordance with MIL-STD-810, method 509, procedure I. There shall be no evidence of external corrosion or deterioration that might affect subsequent operation. The scale errors after this test shall not exceed the tolerances specified for room temperature scale error.

4.6.19. Radio noise interference. The indicator shall be subjected to a radio noise interference test in accordance with MIL-STD-461 except that the frequency range for radiated noise shall be within 0.15 to 150 megacycles. During this test, the synchro shall be operated at 30 ±5 rpm.

4.6.20 Unit torque gradient. The indicator shall be connected to the master transmitter synchro and the master transmitter locked in the electrical zero position. The synchro rotors shall then be energized with rated voltage and frequency ±1 percent. Weights shall be applied to a torque pulley which is positioned on the rotor shaft to rotate the shaft in a ccw direction through an angle of 5° of arc. The torque tending to return the shaft to its original position shall be computed and shall be no less than 2,000 milligram-millimeters per degree of displacement.

4.6.21 Lighting tests. Lighting tests shall be conducted in accordance with the sampling plan B tests of MIL-L-27160, except the dielectric test shall not be conducted.

4.6.22 Reliability. Seven indicators shall be tested for reliability. Acceptance permitting production and total test time shall be in accordance with MIL-STD-781, test plan I. Recording, data handling, and reporting shall also be in accordance with MIL-STD-781.

4.6.22.1 Test. The test equipment and fixtures shall include provisions for vibration and temperature cycling at the periodic intervals specified

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herein. During this test, input signals shall be applied to the indicator at a rate of 2 to 3 cycles (minimum and maximum) per minute from 0 to 85 percent of full scale and return. The test shall begin when rated power is applied. After completion of 4 hours of operation with rated power on, which shall include at least 480 cycles of operation, the power shall be turned off and shall remain off for a period of 10 minutes. For at least 10 minutes out of every hour of ON time, the indicator shall be vibrated in accordance with test level J of MIL-STD-781. The temperature cycles shall be conducted in the following sequence: 4 hours at -20°C, 14 minutes for temperature change to +50°C at a minimum rate of 5°C per minute; 4 hours at +50°C, 14 minutes for temperature change to -20°C, and 4 hours at -20°C. In case of malfunction, the malfunction shall be recorded and the test interrupted while repairs are made. Upon completion of each 250 hours of cycling, the indicator shall be checked for scale error. The scale error reading shall not exceed 150 percent of the room temperature tolerance in table II. No recalibration shall be performed on the units during this test. The test may be interrupted for weekend shutdown. The contractor shall prepare a report in accordance with MIL-STD-781 which shall contain test procedures, details of any malfunctions and corrective actions taken.

4.6.23 Longevity. At the completion of the reliability test, testing of three of the indicators shall be continued for a total of 3,000 hours each. The test time accumulated during the reliability test shall be applied to this test. The accept-reject criteria is not applicable.

4.7. Inspection and testing of preservation, packaging, packing, and marking. The inspection and testing of preservation, packaging, packing, and marking shall be in accordance with MIL-P-116 to insure compliance with section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C", as specified. (see 6.2)

5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in quantity unit pack (QUP) of one each. Each item will be provided a preservation method IC-1 IAW MIL-P-116.

5.1.2 CLEANING. Item shall be cleaned IAW MIL-P-116 C-1 process.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved IAW MIL-P-116 and MIL-STD-2073-1 to provide a method IC-1.

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5.1.5 Level C. Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073-1.

5.2 Packing. Packing shall be level "A", "B", or "C", as specified (see 6.2)

5.2.1. Container-cushioning. Unit container requirements shall conform to MIL-B-121, barrier material, unless otherwise specified by the contracting activity. Unit containers shall be large enough to allow for application of sufficient cushioning/wrap material, between the item and unit container. Item cushioning/wrap material shall conform to PPP-C-795 and provide sufficient density and thickness to protect the unit container from punctures. Intermediate container requirements shall conform to PPP-B-636, weather-resistant, unless otherwise specified by the contracting activity. Intermediate containers shall be large enough to allow for application of sufficient cushioning material, between the unit container and intermediate container. Cushioning material shall be of sufficient density and thickness to insure shock transmission does not exceed peak values in G's established for the item.

5.2.2 Level A. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PPP-B-636, weather-resistant, unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size and minimum tare and cube, consistent with the protection required. Closure shall be in accordance with appropriate PPP-B-636 procedures, as specified by contractor, special packaging instructions (SPI), and contracting activity.

5.2.3 Level B. Same as Level "A" unless otherwise specified by special packaging instructions (SPI) or the contracting activity.

5.2.4 Level C. Item will be packed in such a manner that will afford adequate protection against physical damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to MIL-STD-2073-1.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked IAW special markings required by the contractor, special packaging instructions (SPI) and MIL-STD-129.

5.4 Inspection and test. Test of methods of preservation shall be accomplished in accordance with section 4 of MIL-P-116 to insure compliance with section 5 of this specification. Packaging tests shall be conducted IAW rough handling as specified in FED-STD-101.

6. NOTES

(This section contains information of a general or explanatory nature

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that may be helpful, but not mandatory)

6.1 Intended use. The indicators covered by this specification are intended for use in conjunction with the TRU-80/A or the MS28005-2 oil pressure transmitter in indicating remotely the oil pressure in aircraft systems which require integrally lighted instruments.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Information, if any, to be submitted with the qualification samples

c. When sampling plan B tests will not be conducted

d. Levels of packaging and packing required and reinspection date markings (see section 5).

e. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the OC-ALC/TICLA, Tinker AFB OK 73145-5990 and information pertaining to qualification of products may be obtained from that activity.

6.4 Subject term (key word) listing.

EGU-2A
Oil pressure indicator
Synchro-style

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
AIR FORCE-99

Preparing Activity:
AIR FORCE-71

Project Number:
6620-F503

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-27188B(USAF)	2. DOCUMENT DATE (YYMMDD) 94/04/30

3. DOCUMENT TITLE
INDICATOR, PRESSURE EGU-2/A, OIL, 0-100 PSI

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME OC-ALC/TICLA	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code) TINKER AFB OH 73145-5990	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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